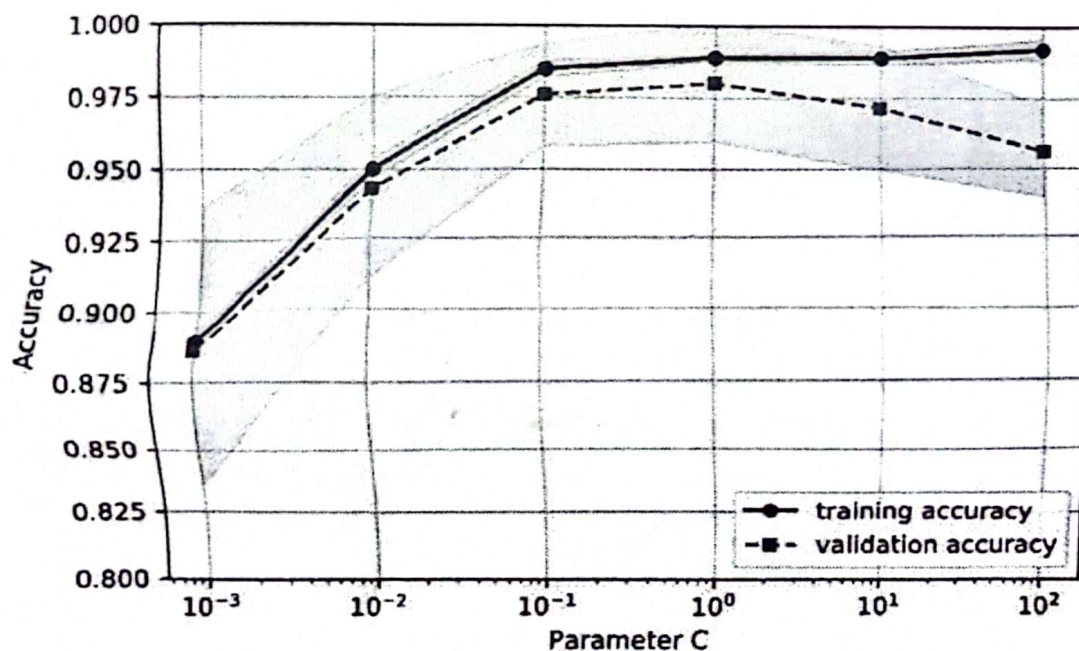


CSE 475, T#2, Time: 30 min, Marks: 20 [= 5x4]

1. How do you perform one-hot encoding on nominal features? What is the disadvantage of this coding?
2. How is L_1 regularization used for feature selection? Explain.
3. What is the cost function of the logistic regression method? Plot the cost for single-instance at different values of $\phi(z)$.
4. Explain the curves in the following plot.



	Target	Prediction
1	Malignant ↙	Not Malignant
2	Malignant ↙	Malignant
3	Not Malignant	Not Malignant
4	Malignant ↙	Malignant
5	Not Malignant	Malignant
6	Not Malignant	Not Malignant
7	Not Malignant	Not Malignant
8	Malignant ↙	Malignant
9	Malignant ↙	Not Malignant
10	Not Malignant	Not Malignant

5. Why is nested cross-validation used? Explain with a figure.
6. Suppose you have written a classifier for diagnosing if a tumor benign or malignant. You have tested your classifier with some data and got predictions as above.
Compute the confusion matrix from the output. Calculate Accuracy, Precision, and Recall from that matrix.